

Radiomics and Deep Learning for Tumor Diagnosis and Prognosis

Abstract

Radiomics and deep learning have become two of the most influential computational paradigms in oncologic imaging. Radiomics converts routine medical images into high-dimensional quantitative descriptors of tumor phenotype, whereas deep learning learns hierarchical image representations directly from data. Over the last decade, these approaches have moved from proof-of-concept studies toward increasingly ambitious systems for lesion detection, differential diagnosis, grading, molecular subtyping, response assessment, recurrence prediction, and survival modeling across brain, lung, head and neck, breast, prostate, gynecologic, hepatobiliary, pancreatic, and colorectal cancers. This survey reviews the methodological foundations, representative applications, and translational barriers of the field, with emphasis on the relationship between handcrafted radiomics, end-to-end deep models, hybrid feature fusion, multimodal learning, and radiogenomics. A central conclusion is that the field has matured beyond the question of whether imaging contains prognostic or diagnostic signal; the more consequential questions now concern robustness, transportability, calibration, causal relevance, and clinical utility. Handcrafted radiomics remains analytically attractive because feature definitions can be standardized and biologically interrogated, but its sensitivity to image acquisition, segmentation, and preprocessing continues to limit reproducibility. Deep learning can learn richer task-specific representations and often improves predictive performance, yet it is even more vulnerable to data hunger, shortcut learning, hidden stratification, and brittle external generalization. Hybrid systems that combine handcrafted features, deep embeddings, and clinical variables currently offer one of the most pragmatic routes to clinical deployment. The survey also reviews emerging directions, including self-supervised and foundation-model pretraining, promptable segmentation, federated learning, vision-language modeling, and multimodal survival modeling. Across the literature, the strongest evidence supports a future in which quantitative imaging biomarkers are not used as stand-alone oracles, but as calibrated, externally validated, workflow-aware components of broader precision-oncology systems. ¹

Introduction

Radiomics was introduced as a framework for extracting quantitative descriptors from standard-of-care images and relating them to diagnosis, prognosis, and treatment response. Its early conceptual appeal was straightforward: tumors are spatially heterogeneous biological systems, and medical images already capture that heterogeneity at scale, noninvasively, and longitudinally. Foundational work by Lambin, Kumar, Aerts, Gillies, and others established the core thesis of the field: intensity distributions, shape, texture, wavelet decompositions, and peritumoral patterns can encode clinically meaningful phenotypes not visible to the naked eye [1–14]. At the same time, this literature revealed the essential technical fragility of the pipeline. Segmentation strategy, voxel spacing, gray-level discretization, reconstruction kernel, noise, contrast timing, and feature software all influence the stability of radiomic descriptors [8–20]. Standardization efforts, especially the Image Biomarker Standardisation Initiative, converted what had been a loosely defined practice into a more formal quantitative discipline [9, 12, 13]. ²

Deep learning entered oncologic imaging from a different direction. Rather than defining features a priori, convolutional and later transformer-based networks learn task-specific representations from raw or minimally processed data. In tumor imaging, deep learning promised to reduce the hand-engineering burden of radiomics, exploit fine-grained spatial context, incorporate three-dimensional appearance, and unify detection, segmentation, classification, and prognosis in a single trainable framework [21–30]. The field rapidly diversified: CNNs for nodule classification, U-Net variants for tumor segmentation, residual networks for molecular prediction, transformers for volumetric context modeling, and increasingly multimodal systems for combining radiology with pathology, genomics, and clinical covariates [21–30, 41–45]. Many recent surveys now treat radiomics and deep learning not as competing schools, but as partially overlapping design spaces in quantitative oncology [31–45]. ³

This convergence has been clinically productive. In neuro-oncology, image models have been developed for glioma grading, IDH status, MGMT methylation surrogacy, recurrence mapping, and survival stratification [46–69]. In thoracic oncology, quantitative imaging supports screening, risk stratification of pulmonary nodules, histologic characterization, mutation prediction, treatment-response assessment, and survival modeling [70–89]. Head and neck imaging studies emphasize HPV status, nodal disease, locoregional control, and recurrence [90–102]. Breast imaging focuses on receptor status, HER2 expression, Ki-67, pathologic complete response, and recurrence risk [103–118]. Prostate and other urologic cancers emphasize clinically significant disease, grade group, extracapsular extension, biochemical recurrence, and metastatic progression [119–130]. Gynecologic, hepatobiliary, pancreatic, and colorectal cancer studies increasingly target recurrence, lymphovascular invasion, treatment response, MSI status, and survival [131–180]. Parallel work in radiogenomics and multimodal learning seeks to connect imaging signatures to molecular pathways and to integrate imaging with whole-slide pathology, transcriptomics, and longitudinal clinical data [181–210]. ⁴

The central challenge is no longer the absence of positive results. Thousands of radiomics and deep-learning papers report statistically significant or numerically impressive associations. The problem is that predictive success in retrospective, single-center, internally validated cohorts is not the same as robust and clinically actionable decision support. Recent reporting and quality frameworks—including CLAIM, TRIPOD+AI, PROBAST+AI, CONSORT-AI, SPIRIT-AI, and DECIDE-AI—reflect the recognition that transparency, calibration, bias assessment, external validation, and real-world evaluation are indispensable for translation [22–27, 196–202]. Many modern reviews reach a consistent conclusion: the field is rich in signal discovery but comparatively poor in transportability studies, outcome-impact analyses, and deployment-ready evidence [34–42, 58, 78, 92, 109, 121, 147, 166, 196–202]. ⁵

Within that context, the goal of this survey is to synthesize the literature on radiomics and deep learning for tumor diagnosis and prognosis as a coherent methodological and translational field. The emphasis is not only on what tasks have been attempted, but on what methodological choices repeatedly explain success or failure: region definition, feature stability, fusion design, censoring-aware learning, calibration, external validation, biologic interpretability, and workflow fit. This is best understood as a survey of quantitative imaging biomarkers in oncology, spanning handcrafted radiomics, deep radiomics, hybrid pipelines, multimodal prediction, and emerging foundation-model paradigms. ⁶

Methods

Handcrafted radiomics pipelines. The canonical radiomics pipeline consists of image acquisition, tumor or habitat segmentation, preprocessing, feature extraction, feature reduction, model training, and validation

[1–20]. Feature families typically include first-order histogram statistics, shape descriptors, gray-level co-occurrence, run-length, size-zone, neighborhood dependence, filter-bank, and wavelet features. This representation is attractive because each step is explicit and, at least in principle, auditable. In oncology, handcrafted radiomics has been applied to benign-versus-malignant discrimination, histologic stratification, molecular subtyping, nodal staging, treatment-response prediction, and survival estimation across nearly every major solid tumor class [31–40, 46–69, 70–89, 90–180]. The main methodological strengths of handcrafted radiomics are data efficiency, compatibility with classical statistical learning, and easier post hoc biological interpretation. Its enduring weaknesses are susceptibility to acquisition shift, segmentation perturbation, multiple-hypothesis bias, and instability under seemingly minor preprocessing changes [8–20]. Robust studies therefore favor harmonized reconstruction settings, IBSI-compliant software, perturbation testing, test-retest analysis, nested feature selection, clinical-variable baselines, and external validation [8–20, 22–27]. 7

Deep learning for tumor representation learning. Deep learning reparameterizes the imaging problem. Instead of compressing images into predefined radiomic descriptors, CNNs and transformers learn latent representations optimized for a downstream objective such as classification, segmentation, time-to-event prediction, or multimodal fusion [21–30]. The earliest tumor-imaging models frequently adapted generic computer-vision architectures, but modern oncology systems increasingly exploit 3D U-Nets, residual volumetric encoders, transformer decoders, and self-supervised pretraining tailored to medical image geometry [23–30, 43–45]. In diagnosis tasks, deep models often outperform traditional radiomics when large, diverse training sets are available, especially when the imaging phenotype is subtle and spatially distributed. In prognosis, however, gains are less uniform. Survival endpoints are noisier than diagnosis labels, treatment heterogeneity introduces confounding, censoring complicates optimization, and pure end-to-end learning can be brittle in small cohorts. As a result, many of the strongest studies do not use deep learning as a stand-alone replacement for radiomics, but as a representation learner whose embeddings are subsequently combined with handcrafted features and clinical variables [31–45, 73, 79, 80, 107, 108, 156, 174, 186–191]. 8

Hybrid radiomics and deep radiomics. One of the most durable patterns in the literature is that hybrid systems often outperform either handcrafted radiomics or deep learning alone. In such systems, handcrafted features contribute low-sample-stability and interpretable morphology/texture descriptors, while deep embeddings contribute hierarchical pattern representations and broader contextual cues. Hybridization takes several forms: late fusion of independent model scores, concatenation of handcrafted and learned feature vectors, multitask training with shared backbones, radiomics-guided attention, and radiomics-clinical nomograms augmented with deep features. The attraction of hybrid design is practical rather than ideological. In many studies, deep learning alone is underconstrained by limited data, whereas handcrafted radiomics alone cannot exploit full volumetric context or weak label structure. The hybrid regime therefore functions as a regularized middle ground, particularly in brain, lung, breast, and rectal cancer studies [31, 34, 40, 49, 56, 73, 79, 95, 107, 117, 156, 174, 177, 186]. Hybrid systems also facilitate ablations that clarify what signal is actually being added by deep models. In several domains, the incremental benefit of deep features is real but modest once segmentation quality, clinical staging, and high-quality radiomics are already included. That finding is methodologically important because it argues for stronger baseline construction, not merely larger neural networks. 9

Segmentation, region definition, and habitat modeling. Tumor-region definition remains a decisive modeling choice. Many early radiomics studies used manually outlined gross tumor volumes, but this creates inter-reader variability and weakens scalability. Deep auto-segmentation, atlas-guided contouring,

promptable segmentation, and weak supervision now reduce annotation cost and improve consistency [23, 27–30, 43]. Yet the deeper conceptual issue is not merely who delineates the region, but what region should be delineated. Prognostic information may lie in intratumoral habitats, necrotic cores, enhancing rims, edema, lymph nodes, peritumoral stroma, or organ-level microenvironment rather than in the primary lesion alone. This is visible in glioma work using edema and infiltrative margins [48–69], breast studies modeling peritumoral tissue and recurrence risk [103–118], lung studies incorporating tumor-organ interaction or treatment fields [76–81], and rectal/HCC studies leveraging peritumoral context for metastasis or recurrence prediction [147–180]. Delta-radiomics and longitudinal feature trajectories extend this logic over time by modeling dynamic change under therapy rather than static baseline appearance. Such designs are clinically appealing because many treatment decisions depend on response kinetics rather than baseline morphology alone [77, 153, 177]. ¹⁰

Diagnosis-oriented modeling. Diagnostic tasks span lesion detection, benign-malignant discrimination, tumor grading, molecular subtyping, nodal staging, and prediction of actionable genomic markers. The strongest diagnostic literature tends to occur in settings where conventional imaging leaves meaningful uncertainty and where pathology or molecular assays are invasive, costly, or spatially limited. Glioma studies illustrate this clearly: models for IDH, ATRX, 1p/19q, and recurrence-vs-radiation injury aim to recover molecular or pathophysiologic information noninvasively [46–69]. Thoracic studies target nodule malignancy, EGFR status, pathologic response, and risk stratification [70–89]. Head and neck work focuses on HPV association and occult nodal disease [90–102]. Breast imaging models predict ER/PR, HER2, HER2-low, Ki-67, molecular subtype, and recurrence-score surrogates [103–118]. Prostate models target clinically significant disease, grade group, extracapsular extension, lymph node invasion, and biochemical recurrence [119–130]. Gynecologic and GI applications include myometrial invasion, lymphovascular space invasion, MVI in HCC, MSI in colorectal cancer, and early pancreatic cancer detection [131–180]. Across these tasks, externally validated models are still a minority, but the most credible evidence suggests that quantitative imaging can reliably enrich clinical suspicion and triage invasive testing in selected scenarios [46–180]. ¹¹

Prognosis, treatment response, and survival modeling. Prognostic modeling is methodologically harder than diagnosis because it requires handling censoring, competing risks, treatment heterogeneity, and the possibility that imaging biomarkers are proxies for care patterns rather than tumor biology. Classical radiomics studies often used Cox models or risk-group stratification pipelines with concordance index, Kaplan–Meier separation, and time-independent AUCs. More recent work increasingly uses discrete-time survival objectives, deep survival networks, and multimodal hazard models [189–192]. Prognosis endpoints include overall survival, progression-free survival, locoregional control, recurrence-free survival, distant metastasis, pathologic complete response, and treatment-specific outcomes such as post-TACE or post-immunotherapy benefit [48–69, 70–89, 90–118, 131–180, 186–191]. In glioma, survival risk scores are often fused with age, extent of resection, and molecular markers [49, 52–69]. In lung cancer, studies span screening-derived mortality prediction, SBRT outcomes, long-term survival, and molecularly linked prognostic phenotypes [70–89]. Head and neck work emphasizes recurrence and de-escalation-sensitive risk assessment [90–102]. Breast studies address neoadjuvant response and recurrence surrogates [103–118]. Hepatobiliary and rectal cancer studies increasingly focus on recurrence after surgery, TACE response, distant metastasis after neoadjuvant therapy, and treatment-path selection [147–180]. The best prognostic models are those that report calibration, compare against clinicopathologic baselines, perform external validation, and analyze clinical net benefit rather than discrimination alone [189–202]. ¹²

Tumor-specific patterns across the literature. Certain application patterns recur across anatomies. Neuro-oncology has been an ideal substrate for radiomics because multiparametric MRI, public BraTS data,

and molecularly stratified glioma classification create fertile conditions for method development [46–69]. Lung cancer remains the most mature non-neuro domain because CT is ubiquitous, screening is expanding, and outcomes such as mutation status, pathologic response, and survival are clinically important [70–89]. Head and neck cancer has generated valuable radiomics work around HPV and PET/CT heterogeneity, though reproducibility challenges remain prominent [90–102]. Breast imaging benefits from richly phenotyped receptor labels and neoadjuvant treatment endpoints [103–118]. In prostate cancer, multiparametric MRI has enabled radiomics to target clinically significant disease and extension beyond the gland [119–130]. HCC and pancreatic cancer illustrate the appeal of noninvasive biomarkers where biopsy may be unsafe, limited, or spatially unrepresentative [147–165]. Rectal and broader colorectal cancer form another high-yield domain because baseline MRI/CT and neoadjuvant treatment pathways create repeated opportunities for response prediction and longitudinal risk modeling [166–180]. A survey-level synthesis across these tumor sites suggests that model quality depends less on anatomy per se than on three factors: label quality, imaging standardization, and whether the endpoint is biologically or clinically coherent. ¹³

Multimodal fusion and radiogenomics. The field’s most ambitious direction is multimodal precision oncology. Radiogenomics connects imaging phenotypes to genomics, transcriptomics, epigenetics, or pathway-level biology [181–185]. Radiopathomics and multimodal survival models combine radiology with whole-slide pathology and clinical data [186–188]. These approaches are attractive because imaging provides spatial and temporal coverage, omics provides mechanistic specificity, and pathology provides microscopic confirmation. In practice, however, multimodal fusion introduces substantial engineering and statistical complexity: missing modalities, cohort-size shrinkage, alignment errors, overfitting, and unclear failure attribution. The most convincing multimodal studies therefore tend to use simple but disciplined fusion strategies—cross-attention, modality-specific encoders with late aggregation, or carefully regularized joint embeddings—rather than unconstrained end-to-end architectures. Even when multimodal systems improve prediction, their real scientific value often lies in linking latent imaging patterns to biologically plausible pathways, thereby moving the field from black-box association toward mechanistic hypothesis generation [181–188]. ¹⁴

Foundation models, self-supervision, and data-efficient adaptation. A decisive recent trend is the move from task-specific supervised training to large-scale pretraining, especially self-supervised encoders, foundation segmentation models, and vision-language systems [43–45, 209–210]. These paradigms are especially relevant to oncology because labels such as survival, recurrence, and genomic status are expensive and sparse, whereas raw imaging archives are abundant. Self-supervised pretraining can improve feature quality in low-label regimes; foundation models can reduce annotation burden for segmentation; and vision-language models may eventually support report-aware representation learning and weak supervision [43–45, 209–210]. Still, the translational frontier is wide open. Foundation models trained on broad medical imaging distributions do not automatically solve oncology-specific problems such as subtle scanner shift, rare molecular phenotypes, or treatment-confounded hazards. Their value will likely be greatest when paired with task-specific fine-tuning, uncertainty estimation, and rigorous external evaluation rather than treated as universal plug-and-play solutions. ¹⁵

Challenges and Prospects

The first major challenge is **reproducibility**. Radiomics has spent much of its history rediscovering how sensitive image biomarkers are to acquisition, reconstruction, segmentation, and software implementation [8–20]. The IBSI substantially improved feature definition standardization, but standardization of feature formulas does not eliminate the much broader problem of upstream heterogeneity. Scanner vendor

differences, reconstruction kernels, motion, contrast phases, and institution-specific workflows alter the input distribution before feature extraction or deep inference even begins [9, 16–20]. Harmonization methods such as ComBat can help, but they are not substitutes for biologically coherent cohort design or prospective imaging protocols. The same is true for deep learning: domain shift changes learned representations in ways that are harder to diagnose than handcrafted feature perturbations [193–195, 203–208]. A likely future direction is tighter integration of harmonization, uncertainty modeling, and domain-aware validation during development, rather than treating these as afterthoughts. ¹⁶

The second major challenge is **validation realism**. Many studies still rely on random train-test splits within a single institution, sometimes after weak leakage control or post hoc feature selection. This inflates apparent performance and obscures what happens under geographic, temporal, or workflow shift. Hidden stratification, shortcut learning, and site-specific correlations are especially dangerous in medical imaging because models may exploit scanner signatures, annotation style, treatment-era differences, or prevalence artifacts rather than true tumor biology [11, 193–195]. Oncology models are additionally vulnerable to outcome leakage through treatment variables that are downstream of disease severity. External validation, temporal validation, calibration analysis, and decision-curve or utility-oriented evaluation therefore need to become default expectations, not premium extras. For prognosis, the reporting standard should include not only c-index but also calibration over time, censoring-aware discrimination, and sensitivity analyses for treatment heterogeneity [189–202]. ¹⁷

The third challenge is **biological interpretability**. Enthusiasm for high-performance prediction has at times outpaced understanding of what imaging signatures actually represent. Radiomics is often described as more interpretable than deep learning, but that is only partially true. A feature called entropy or gray-level nonuniformity is interpretable at the level of algorithmic definition, not automatically at the level of pathobiology. Conversely, a deep feature can be biologically informative even if it lacks a human-readable name. The literature is therefore moving toward radiogenomics, radiopathomics, saliency-grounded analysis, feature clustering, and pathway enrichment to establish whether imaging signatures map to meaningful phenotypes such as necrosis, angiogenesis, hypoxia, proliferation, immune state, or stromal architecture [181–188]. The most promising future work will likely treat interpretability as a multiscale alignment problem: image-to-region, image-to-pathology, and image-to-omics, rather than as a choice between “handcrafted interpretable” and “deep black box.” ¹⁸

The fourth challenge is **clinical integration**. Even a well-validated model may fail if it does not fit workflow, intervention timing, reader trust, or care pathways. A useful pancreatic-detection model must influence triage soon enough to matter; a glioma survival model must add information beyond molecular pathology; a breast-response model must alter neoadjuvant decision-making without increasing harm; a rectal-cancer model must be coordinated with imaging schedules and multidisciplinary review. Reporting frameworks such as CLAIM, TRIPOD+AI, CONSORT-AI, SPIRIT-AI, and DECIDE-AI are important because they move the field from “Can a model predict?” to “How was it developed, validated, and used in the real clinical setting?” [22–27, 197–202]. Future studies should be designed around decision points, not just endpoints. This means prespecifying intended users, comparator pathways, threshold selection, alert burden, uncertainty handling, and downstream actions. ¹⁹

The fifth challenge is **privacy, equity, and data scale**. Oncology imaging cohorts are often fragmented across institutions, each too small to support robust deep learning for rare endpoints. Federated learning offers a plausible alternative by enabling collaborative training without centralizing raw data [203–208]. Yet federated learning does not automatically solve bias, label inconsistency, or imaging heterogeneity; in some

settings it can even amplify them if local datasets are highly imbalanced. What it does offer is a governance-compatible mechanism for scaling partnerships and testing external robustness during model development. Combined with self-supervised pretraining and foundation-model adaptation, federated oncology imaging could materially reduce the small-sample bottleneck that has constrained both deep learning and radiomics. Still, privacy-preserving scale should not be confused with fairness. The field will need routine subgroup analysis, demographic calibration checks, and site-specific accountability if these systems are to be trusted in practice. ²⁰

The most important prospect is that **radiomics and deep learning are converging into a broader quantitative-oncology stack**. In this stack, segmentation, representation learning, handcrafted features, pathology fusion, genomics integration, survival modeling, and workflow evaluation are all modular components. The next generation of impactful studies will likely share several design principles: prospectively curated multicenter cohorts; explicit train-validation-test-time separation by institution or era; robust baseline comparison with standard clinicopathologic models; uncertainty-aware prediction; biological alignment through radiogenomics or pathology; and deployment-oriented evaluation. Technical novelty alone is becoming a weaker differentiator. The strongest future contributions will be those that show a model is stable, calibrated, transportable, interpretable enough for its use case, and actionable at a clinically meaningful decision point. ²¹

Conclusion

Radiomics and deep learning have fundamentally changed the intellectual landscape of oncologic imaging. The literature now provides strong evidence that routine images contain rich information relevant to tumor diagnosis, molecular characterization, treatment response, recurrence, and survival. Across cancers, handcrafted radiomics has demonstrated the enduring value of explicit quantitative image biomarkers, while deep learning has expanded the representational capacity of image-based oncology. The most mature lesson of the field is that neither paradigm is universally superior. Handcrafted radiomics excels in transparency and data efficiency but is fragile to technical variability; deep learning can learn richer patterns but requires stronger data curation, broader validation, and more careful failure analysis. Hybrid and multimodal systems currently offer the best compromise. Going forward, the decisive issues are not marginal gains on retrospective leaderboards, but reproducibility, calibration, biologic grounding, external transportability, and integration into real clinical decisions. If the field meets those requirements, radiomics and deep learning will not remain ancillary research tools; they will become core components of precision oncology. ²²

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